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1 Spatial dynamic rupture – late sliver-seeded V_max blow-up (job 7748818, 2026-05-25)

Third documented SAFS dynamic-rupture blow-up. **Slip-weakening (LSW)**, not rate-and-state. Distinct in *timing and location* from the two prior events but shares the **mesh-sliver** root cause with the 2026-05-20 north-tip event.

	this event (2026-05-25)	2026-05-22 speckle	2026-05-20 north-tip
job	7748818	7744292	(postevent)
config	..._slip_weakening_safs_	projected_safs_	slip_weakening_safs_
onset	late, t≈28 s (runaway t≈42 s)	early, t≈0.84 s (nucleation)	post-event, t≈3 s

	this event (2026-05-25)	2026-05-22 speckle	2026-05-20 north-tip
where	mid-fault $x \approx 365$ km + shallow trace $x \approx 535$ km	on/next to nucleation patch	far north tip
recovers?	NO $\rightarrow$ 5e64, never recovers	yes $\rightarrow$ back to ~ 3 m/s	no $\rightarrow$ NaN
root cause	mesh slivers (this doc)	self-limiting on-patch speckle	mesh slivers (shallow trace)

Correction to the 2026-05-22 doc. That doc states the 500m `z0embed` mesh is “sliver-free.” **It is not.** This investigation found the single worst triangle in the entire fault mesh (min-angle **9.1°**, `tri_q` **0.245**) at $x \approx 365$ km, plus a sliver band at the shallow surface trace. The 05-22 *speckle* was indeed not sliver-driven (it was on the well-meshed patch and recovered); but the mesh as a whole still carries sliver patches, and **this** blow-up sits squarely on them.

Run header (from `/Users/chunhuizhao/Documents/spatial_dyn_resDc2_7748818.log`):

```
config: safs/.../spatial_friction_slip_weakening_safs_projected_stress_resolution_Dc2.toml
mesh:   safs/.../meshing/results/msh/safs_fault_box_nwcut_500m_lcfar3000_z0embed.msh
law:    slip_weakening      ader order: 2      cfl: 0.5      use pml: no
tfinal: 100 s              ranks: 500        num_fault_global = 128151
[time]  dt = 0.000349884 s  nsteps = 285809
[derived] mesh h_min = 197.029 m (scalar cp)
[derived] L_nuc/h_min min/max = 6.09 / 19.45 (NB: should be >= 10) <-- under-resolved warned
[derived] PASS: initial conditions are well-posed.
```

1.1 1. Symptom – late, super-exponential, non-recovering

The rupture nucleates and propagates **correctly** for ~ 25 s, then a localized instability grows and runs away:

t (s)	step	V_max (m/s)	regime
0–1.75	0–5k	0 $\rightarrow$ 4	nucleation ramp (gradual overstress)
~ 3	8.4k	~ 10	rupture breaks out, propagates
3 $\rightarrow$ 28	8k–80k	stable 10–12	healthy front; <code>n_rupturing</code> 3.6k $\rightarrow$ 8.8k; survives the 6.94 s reflection window

t (s)	step	V_max (m/s)	regime
28	80k	15	first departure from the plateau
33 / 42	94k/120k	30 / 100	slow climb
44 / 46	125k/132k	1e3 / 1e6	super-exponential cascade
64.6	184.6k	5e64	fully blown up; n_rupturing=113k/128k (fault-wide)

The LSW physics, fault flux, and nucleation are therefore **correct** — a clean ~ 10 m/s rupture for ~ 25 s. The blow-up is a **late, growing numerical instability**, not a setup error.

1.2 2. Trace – normal-stress “chattering” (fault open/close), not physical slip

Tracing the dominant runaway DOF (365100, 3.83892e6, $z=-2488$ m) through the cascade ([DIAG-ONSET] records, SEAS_DIAG_BLOWUP=1):

```
V=749   sigma_n_corr=-7.15e8   tau2_corr=-2.3e-7   <- tensile sigma_n, traction ~ 0 (fault OPEN)
V=1027  sigma_n_corr=+1.39e8   tau2_corr=-6.6e7    <- compressive again, traction back
V=19896 sigma_n_corr=-2.6e10   tau2_corr=-7.9e-6
V=53044 sigma_n_corr=+9.2e10   ...
```

sigma_n_corr **flips sign every step (tensile $\leftrightarrow$ compressive) with exponentially growing amplitude**; whenever it goes tensile, the Godunov fault flux opens the fault and the shear traction collapses to $\sim 0 \rightarrow$ no restoring friction $\rightarrow V$ runs away. This is a grid-scale **normal-stress oscillation**, the same [[v_n]] class noted in project_safs_vn_leak_not_frame (per-sub-step predictor/ghost, R-1303/R-1601) — here amplified to destruction.

1.3 2c. Field-data confirmation (fault.vtkhdf, 2026-05-26) – tensile σ_n + free slip CONFIRMED

The full fault output (/Users/chunhuizhao/Documents/fault.vtkhdf, 10.4 GB, 1301 snapshots $t \in [0, 64.59]$ s, 128 151 points) lets us verify the σ_n -chatter / fault-opening mechanism from the *saved fields* (normal_stress, traction_{strike,dip}, slip_rate_{strike,dip}), independent of the log. Figure: blowup_clusterA_sigma_n_tau_V_2026-05-25.png.

Cluster A point (idx 114460, x365029 y3838938 $z=-2560$ m) trajectory:

t [s]	σ_n [Pa]	$ \tau $ [Pa]	$ V $ [m/s]	$\tau/ \sigma_n $	state
24–43	+8.42e7	2.56e7	0	0.304	locked at the μ_d threshold

t [s]	σ_n [Pa]	$ \tau $ [Pa]	$ V $ [m/s]	$\tau/ \sigma_n $	state
44.5	+5.69e7	3.29e7	1.88	0.58	instability arrives; τ overshoots
45.0	+1.02e9	6.3e7	275	0.06	σ_n spikes $20\times$ (oscillation explodes)
45.4	−2.91e9	1.77e9	1744	0.61	first TENSILE — fault opens
45.6	−3.23e10	1.35e9	1.8e4	0.04	tensile $\rightarrow \tau$ collapsing (free slip)
46.0	−4.70e11	1.96e11	1.25e5	0.42	super-exponential cascade
64.0	−1.27e69	1.82e66	5.76e62	0.001	tensile, $\tau \rightarrow 0$ (free slip)

Both hypotheses confirmed: - **Tensile normal stress — YES.** σ_n at the blow-up points goes strongly **negative** (compression is +; first tensile at $t \approx 45.3$ s; 199/1301 snapshots tensile), reaching **−1.3e69 Pa**. The healthy reference point (nucleation patch, idx 62662) stays **compressive the whole run** ($\sigma_n \in [49.5, 59.5]$ MPa, 0/1301 tensile, $|V|_{\max}$ 3.6 m/s) — the contrast is decisive. - **Free slip ($\tau \rightarrow 0$) — YES, on the tensile half-cycles.** When σ_n is tensile the fault opens and $\tau/|\sigma_n|$ collapses to **0.001–0.04** (vs $\mu_d=0.30$) — it sheds shear restoring traction. It is a **normal-stress chatter** (σ_n alternates tensile-open $\leftrightarrow$ compressive-at- μ_d each macro-step; the 0.05 s snapshots alias the per-step flips the log showed), so the median $\tau/|\sigma_n|$ over the whole runaway (0.33–0.39) is the mixture of free-slip (~ 0) and at-limit (~ 0.30) half-cycles.

Sequence (mechanism). A locked patch ($V=0$, $\tau/\sigma_n=0.30$) $\rightarrow$ the slowly-growing instability arrives ($\sim t=44.5$ s) $\rightarrow$ the σ_n oscillation amplitude explodes over ~ 5 macro-steps (84 MPa $\rightarrow$ 1 GPa) $\rightarrow \sigma_n$ crosses **tensile** ($\sim t=45.4$ s) $\rightarrow$ fault **opens**, $\tau \rightarrow 0$ (free slip) $\rightarrow$ no restoring force $\rightarrow$ super-exponential $|V|$ runaway. The Godunov fault flux *correctly* opens the fault under tension (faults can't sustain tension) — but here the tension is **numerical** (the spurious oscillation), so the opening converts a normal-stress wiggle into a real loss of shear restoring $\rightarrow$ positive feedback. **This is the runaway stage; the σ_n oscillation that grows it ($t=28$ – 45 s) is the spatial-operator instability of §6b (central-flux suspect).**

Spatial spread (snapshot counts, of 128 151 points): tensile $\sigma_n / |V| > 50$ m/s grow from **801 / 0** at $t=40$ s $\rightarrow$ **1179 / 39** at $t=46$ s $\rightarrow$ **12 780 / 14 682** at $t=52$ s $\rightarrow$ **48 343 / 111 574** at $t=64.6$ s. So it seeds at ~ 800 sliver/near-surface points and the opening+radiation destabilizes neighbours until it is fault-wide — matching the log's $n_{\text{rupturing}} \rightarrow 113$ k.

1.4 3. Where – two clusters, both on low-quality (sliver) fault elements

[DIAG-ONSET] logs the **global argmax DOF per step** (printed only when $V_{\max} > 10$). Aggregating peak $|V|$ and argmax-hit-count per location gives 9 runaway hotspots (peak $|V| > 50$ m/s; physical peak was ~ 10 – 15), in **two tight clusters, both far from the nucleation patch** ($x \approx 607$ km, $z = -5.1$ km):

cluster	x	y	z	peak $ V $ (m/s)	argmax hits
A (mid-fault)	365 040–365 100	3.8389e6	–2.5 to –2.7 km	up to 8.9e64	31339 + 17258 + ...
B (shallow trace)	535 056–535 364	3.7553e6	–25 to –122 m	up to 7980	8799 + 2527 + ...

(Full list: `.../meshing/results/vtu/safs_fault_blowup_points.csv`.)

1.5 4. ROOT CAUSE – the blow-up elements ARE mesh slivers

`meshing/code/locate_fault_slivers.py` + a fault-wide quality sweep on `safs_fault_box_nwcut_500m_lcfar3000` (fault physical tag 101, 42 358 triangles):

metric	fault-wide median	fault-wide worst	the 20 marked blow-up cells
min angle	58.9°	9.10°	median 32.8°, min 14.0°
tri_q (1 = equilateral)	0.9995	0.245	median 0.795, min 0.312
min edge aspect	499 m 1.02	112 m 4.43	down to 112 m up to 3.86

Correlation: - The marked cells' **median** min-angle (32.8°) is at the **0.80th percentile** fault-wide — worse than 99.2 % of all fault triangles. - The **single worst triangle in the whole mesh** (9.10°, tri_q 0.245, centroid (365012, 3838896, –4277)) is at **cluster A**; the 2nd and 3rd worst (10.6°, 11.8°) are also at cluster A. - Fault-wide only 274/42358 (0.65 %) triangles have min-angle < 30°, yet **8 of the 20** marked cells do — a $\sim 40\times$ enrichment. - Cluster A: min-angle 9–14°, min_edge **112 m**. Cluster B (shallow trace): min-angle 20–25°, min_edge **181 m**. The well-meshed fault is $\sim$ equilateral with 500 m edges.

Mechanism (where × when × how)

The fault is excellent almost everywhere, but sliver patches have an effective element size (min_edge 112–181 m, sliver **inradius** far smaller) well below the $h_{\min} = 197$ m used to set the global explicit dt. Those slivers are the worst-conditioned points on the fault. Sequence: 1. **Where** = slivers. Cluster A is the global-worst triangle; cluster B is the shallow-trace sliver band. Both are exactly

where the runaway localizes. 2. **When** = energy accumulation. The run is stable through the first reflection passes ($t \approx 7\text{--}28$ s); with **no PML** (only 1st-order absorbing BCs on attrs 103/104) and $t_{\text{final}} = 100 \text{ s} \gg 6.94 \text{ s}$ reflection time, reflected-wave energy accumulates and at $t \approx 28$ s the marginally-unstable sliver modes begin to grow. 3. **How** = σ_n chatter. The growing mode drives σ_n tensile $\rightarrow$ fault opens $\rightarrow$ shear traction $\rightarrow 0 \rightarrow$ unbounded slip-rate growth (§2).

Slivers set *where*; accumulated reflection energy + the $\sigma_n/[v_n]$ leak set *when/how*. Cluster A being the literal global-worst sliver, and the dominant runaway, is not coincidence.

This matches `locate_fault_slivers.py`'s own provenance note (written for the 2026-05-20 event): “spurious north-tip velocity coincided with sliver triangles at the shallow fault-surface trace.”

1.6 5. Artifacts produced (2026-05-25)

In `.../meshing/results/vtu/`: - `safs_fault_blowup_marked.vtu` — fault surface (42 358 tris) with cell arrays `is_blowup` (0/1), `blowup_peakV`, `blowup_loghits`, **`min_angle_deg`**, **`tri_q`**, **`aspect`**, **`min_edge_m`**. Color by `min_angle_deg / tri_q`: the marked blow-up cells sit in the dark sliver patches. - `safs_fault_blowup_points.vtu / .csv` — the 982 argmax hotspots; `is_blowup` flags the 9 with peak $|V| > 50$ m/s.

Generators (re-runnable, conda activate `pythonenv`): - `.../meshing/code/mark_blowup_on_fault.py` — `log` $\rightarrow$ marked VTU/points (`VBLOW=50`). - `.../meshing/code/locate_fault_slivers.py` — spatial quality probe (`--target X Y Z --radius R --fault-tag 101`).

Reproduce the cluster-A quality:

```
python3 code/locate_fault_slivers.py \
  results/msh/safs_fault_box_nwcut_500m_lcfar3000_z0embed.msh \
  --target 365100 3838920 -2487.95 --radius 1500
```

1.7 6. Recommended actions (highest leverage first)

1. **Repair the sliver patches** at $x \approx 365$ km ($z = -0.9$ to -4.3 km) and the shallow surface trace ($x \approx 535$ km, $z \approx -0.1$ km). This is the *where*; remove it and the instability has no seed. Plans already exist: `meshing/docs/PLAN_mesh_quality.md`, `PLAN_retriangulate.md`, `EXPLORE_tmap_mesh_optimization.md`. The “`z0embed`” surface-embedding step is the likely sliver source (fault meeting the $z=0$ free surface at a grazing angle $\rightarrow$ thin triangles).
2. **Confirm the local-CFL story on the 3D tets** (TODO). The surface triangle is a proxy; the explicit Δt is set by the *tet* `inradius/cp`. Check the tet quality at cluster A — if a sliver tet's stable $\Delta t < 3.5e-4$ s, that is a true local-CFL violation. (Open: why stable for 28 s then runaway? — accumulated energy, not an immediate hard CFL trip.)
3. **Run inside the valid window** regardless: $t_{\text{final}} \leq \sim 7$ s or wire real PML (the driver's `use_pml` flag is currently a no-op — see `REVIEW.md R-026`). The blow-up is entirely in the already-reflection-contaminated regime ($t > 6.94$ s).
4. **Survey other-code sliver handling** (next task): how `SeisSol` treats degenerate/sliver elements (local time stepping, element dropping, quality gates, CFL per-element). Inform

whether MFEM-SEAS needs per-element dt or a mesh-quality hard gate.

1.8 6b. UPDATE (2026-05-25, later) – it is NOT a time-stepping/CFL problem

Verified ComputeMaxDt (wave_operator.inl:5634) and the tet geometry:

- MFEM-SEAS sets a **single global dt = min over tets of $\text{cfl_eff} \cdot h_e / \text{cp_e}$** , where h_e = the tet **inscribed diameter** ($6V/A_{\text{total}}$, wave_operator.inl:97) — the *same* geometric measure SeisSol uses (GlobalTimestep.cpp:computeCellTimestep, insphere). So MFEM-SEAS is **not** naive; the dt is already sliver-aware.
- Tet inscribed-diameter over the 3.69 M tets: **min 42.0 m**, p1 127 m, median 322 m. The reported $h_{\text{min}} = 197$ m is a *different* metric (`pmesh.GetElementSize(e,0)`), used only for the L_{nuc}/h display) — the **dt actually uses 42 m**: $\text{dt} = (\text{cfl}/9) \cdot 0.9 \cdot 42 / \text{cp} \approx 3.5 \times 10^{-4}$ s matches the log exactly (cfl 0.5, DG 3(2p+1) factor, mixed-flux 0.9).
- The cluster-A blow-up **tets** have min inscribed-diameter **46 m** = $1.10 \times$ the global limiter (42 m). Cluster B = 95 m ($2.27 \times$). **So dt is sized essentially for the very elements that blow up** — they are CFL-stable by construction.

Conclusion: dt is already set by the worst sliver, and they still blow up $\Rightarrow$ this is a SPATIAL-operator instability, not a temporal/CFL one. A CFL-stable dt only guarantees stability if the semi-discrete operator is dissipative/neutral (all eigenvalues ≤ 0). If the DG fault/interior operator has a weakly **positive** eigenvalue on the distorted sliver elements, the mode grows as $\exp(\lambda t)$ for **any** dt — no dt reduction and no LTS fixes it. The “stable 28 s then exp runaway at a CFL-safe dt” signature is exactly a positive-eigenvalue spatial mode fed by accumulating energy, not a CFL trip.

Leading suspect — non-dissipative central flux. This run used `mixed_flux = adjacent`: the log shows **200 308 interior faces on central (non-dissipative) flux** (Zhang 2023). The ComputeMaxDt comment itself flags “central flux is non-dissipative.” Central flux + a 9° sliver’s poor conditioning removes the dissipation that would otherwise damp the $\sigma_n/[v_n]$ leak (`project_safs_vn_leak_not_frame`, R-1303/R-1601 predictor/ghost). SeisSol uses **upwind (Gondunov)** interior flux — dissipative — so its insphere-CFL dt is sufficient; MFEM-SEAS’s central-flux choice removes that safety net on distorted elements.

Reflections set the when, not the root. No PML + $t_{\text{final}} 100 \text{ s} \gg 6.94 \text{ s}$ lets reflected energy accumulate; the unstable mode needs amplitude to become visible (onset $t \approx 28 \text{ s}$). PML / shorter t_{final} would *delay* it, not cure a positive-eigenvalue mode.

Answers to the two design questions

Q1 — sliver standard. The project gate already exists (`check_mesh_quality.py`: Q1 min tet edge ≥ 100 m, Q2 Joe-Liu $\eta > 0.1$). The 500 m mesh **passes** it ($\eta_{\text{min}} 112$ m, $\eta_{\text{min}} 0.184$) yet blows up — so the gate is **too lenient for this scheme**. Define the standard on the quantity that drives each failure mode: (a) **fault triangle** min-angle $\geq \sim 25\text{--}30^\circ$ / $\text{tri_q} \geq \sim 0.5\text{--}0.6$ (the σ_n leak lives on the fault face; the blow-up tris were $9\text{--}25^\circ$); (b) **tet** Joe-Liu η floor raised to $\sim 0.2\text{--}0.3$ (0.1838 is marginal). But note: tightening the mesh gate *reduces the amplifier*; it does not, by itself, make a non-dissipative operator stable. **Q2 — time stepping.** Not the lever. dt is already worst-sliver-limited; reducing it or adding SeisSol-style LTS (which is a *performance* optimization, same insphere-CFL per cluster) will **not** stop a positive-eigenvalue spatial mode. The fixes are spatial: remove the slivers (mesh repair), and/or restore dissipation (`mixed_flux = none` / pure upwind),

and/or penalize the fault-normal $[[v_n]]$ weld.

Discriminating experiment (recommended, from a checkpoint near $t \approx 27$ s, step ~ 78000)

1. Rerun with **mixed_flux = none** (pure upwind). If the blow-up vanishes or moves vastly later $\rightarrow$ central flux is confirmed as the instability source. (Cheapest, highest-information.)
2. Rerun with **PML / tfinal = 6 s**. If onset just delays/weakens $\rightarrow$ reflections are an amplifier, not the root.
3. Rerun on a **sliver-repaired** mesh (TMOP / retriangulate the $x \approx 365k$ + shallow-trace patches) at the same flux. If stable $\rightarrow$ the sliver conditioning was the amplifier.

1.9 7. Open questions

- Is $h_{\min} = 197$ m the bulk-tet min, or does it miss the 112 m sliver edges? (The fault $\min_edge$ $112 < 197$, so $h_{\min}$ does not see the smallest fault feature — worth auditing how dt_cfl is computed.)
- Would the rate-and-state path blow up at the same slivers? (RS regularizes friction but the $\sigma_n / [[v_n]]$ leak is law-agnostic.)
- Does the `--print-derived` gate need a **mesh-quality** check (min-angle / per-element CFL) in addition to $L_{\text{nuc}}/h_{\min}$?

2 Implementation Plan: optional compressive normal-stress strength floor (σ_n cap on shear strength)

STATUS: DRAFT — for /code-review $\rightarrow$ /code-fix $\rightarrow$ /code-implement. Author 2026-05-26. Targets the confirmed free-slip half of the blow-up (§2c): under tension the LSW strength collapses to 0 (SolveLSW_TPV205 clamps σ_n to 0), so the fault free-slides. This adds an **opt-in** floor so the strength stops being proportional to σ_n below a threshold.

2.1 Overview

Expose a **compressive normal-stress floor** σ_{n_floor} (TOML, default = disabled). In every shear-strength computation, replace the normal stress used for strength with $\sigma_{n_strength} = \max(\sigma_{n_total}, \sigma_{n_floor})$. Above the floor the strength is the usual $\mu \cdot \sigma_n$ (proportional); **below the floor it is the constant $\mu \cdot \sigma_{n_floor}$** (no longer proportional to σ_n , and never the spurious tensile 0 (LSW) or $|\sigma_n|$ (RS)). Set $\sigma_{n_floor} = 10$ MPa in the SAFS configs. This breaks the $\sigma_n \rightarrow \text{strength} \rightarrow \text{radiation}$ feedback that drives the runaway (§2c, §6b) while staying **byte-exact** for the TPV/BP5 regressions when disabled.

Governing change (math). For both laws the friction strength is $\tau_{\text{strength}} = \mu_- \cdot \sigma_n^{\{\text{str}\}}$ where the friction coefficient μ_- is unchanged (LSW: $\mu_{\text{eff}}(\delta)$; RS: $f_V = a \cdot \sinh(V \cdot C)$), and

$$\begin{aligned} \sigma_n^{\{\text{str}\}} &= \max(\sigma_{n_total}, \sigma_{n_floor}) && \text{(floor enabled, } \sigma_{n_floor} \geq 0) \\ \sigma_n^{\{\text{str}\}} &= \max(\sigma_{n_total}, 0) && \text{[LSW] / } |\sigma_{n_total}| \text{ [RS] (floor disabled -} \\ &&& \text{current behavior)} \end{aligned}$$

Only the normal stress entering the **strength** is floored; the written-back σ_{n_corr} (the

normal_stress output channel) is untouched. $\sigma_{n_floor} = 10 \text{ MPa} \Rightarrow$ below 10 MPa compression the strength saturates at $\mu \cdot 10 \text{ MPa}$ (a cohesion-like constant), so a spurious tensile excursion can no longer drive free slip.

2.2 Constraints

- **Byte-exact regression is non-negotiable.** TPV205/102/104 and BP5 must be bit-identical when the floor is disabled. $\Rightarrow$ the option is **opt-in via a disabled sentinel** ($\sigma_{n_floor} < 0$), and when disabled each path keeps its *exact* current expression ($LSW \max(\sigma_n, 0)$, $RS |\sigma_n|$). The benchmarks set no floor $\Rightarrow$ untouched.
- **SolveLSW_TPV205 is the byte-exact oracle kernel** (tpv205_friction.hpp). The new parameter MUST default to the current behavior ($\sigma_{n_floor} = 0.0 \Rightarrow \max(\sigma_n, 0.0)$), so the standalone TPV205 driver + the oracle iterator that don't pass it are unchanged.
- **CLAUDE.md “don't change sign conventions / don't revert a fix.”** This is *additive*: the existing $\max(\sigma_n, 0)$ (LSW), $\text{abs}(\sigma_n)$ (RS), and the SEAS_NOOPENING env guard all stay; the floor is a new gated branch. Cite §2c/§6b as justification in the commit.
- **Single carrier.** All four strength sites are reached through one FaultFaceFlux (the iterators bind FaultFaceFlux&), so the floor lives as one FaultFaceFlux member, set once from config; no per-DOF plumbing in v1 (flag per-DOF as a follow-up).
- **Floor is a compressive (positive) stress;** μ_- is the *active* coefficient (continuous at the threshold). See “Decisions for /code-review” for the μ_s -literal alternative.

2.3 Strength sites to change (all four; mapped 2026-05-26)

#	file:line	path	current σ_n in strength
1	dynamic/tpv205_substep_102_104_205.cpp:110	SAFE_LSW (LswFrictionIterator $\rightarrow$ Tpv205) — the blow-up path	via SolveLSW_TPV205 $\rightarrow \max(\sigma_n, 0)$
2	dynamic/fault_face_flux.hpp:804	LSW EvaluateADER_LSW (shared/inline)	via SolveLSW_TPV205 $\rightarrow \max(\sigma_n, 0)$
3	dynamic/fault_face_flux.hpp:804	LSW for 0261-rupture variant	via SolveLSW_TPV205 $\rightarrow \max(\sigma_n, 0)$
4	dynamic/fault_face_flux.hpp:526 and :562	RS substep / shared (aging)	$\text{abs}(\sigma_n)$

2.4 Phase 1 – config plumbing (no physics change)

Goal

A σ_{n_floor} value parses from TOML into the config and reaches FaultFaceFlux; default disabled (sentinel < 0); all existing tests byte-exact.

Files to Modify

- spatial/code/spatial_friction.hpp — add to struct SpatialFrictionConfig: `real_t sigma_n_strength_floor_pa = -1.0; ///< <0 = disabled; >=0 = floor [Pa] on  $\sigma_n$`

in shear strength.

- `spatial/code/spatial_friction.cpp` — in the `[friction]`-level parse (the function that fills `SpatialFrictionConfig`, alongside `law`), read optional key `sigma_n_strength_floor_pa` (default `-1.0`). Validate: if present, ≥ 0 and finite, else `MFEM_ABORT`.
- `dynamic/fault_face_flux.hpp` — add private member `real_t sigma_n_strength_floor_ = -1.0;` and public setter `void SetSigmaNStrengthFloor(real_t v) { sigma_n_strength_floor_ = v; }`.
- `drivers/spatial_dyn_driver.cpp` — after `FaultFaceFlux fault_flux(...)` (:1258): `fault_flux.SetSigmaNStrengthFloor(cfg.sigma_n_strength_floor_pa);`.

Interfaces

- `SpatialFrictionConfig::sigma_n_strength_floor_pa` (`real_t`, default `-1.0`).
- `FaultFaceFlux::SetSigmaNStrengthFloor(real_t)` / member `sigma_n_strength_floor_`.

Edge Cases

- key absent $\rightarrow -1.0$ (disabled). key present and $< 0 \rightarrow \text{MFEM_ABORT}$ (a negative floor is the sentinel, not a value). key = $0 \rightarrow$ allowed ($\equiv \text{LSW } \max(\sigma_n, 0)$; for RS switches $\text{abs} \rightarrow \max(\cdot, 0)$).

Acceptance Criteria

- ☐ New unit test `test_spatial_friction_config`: a TOML with `sigma_n_strength_floor_pa = 10.0e6` parses to `10e6`; absent $\rightarrow -1.0$; negative value aborts.
- ☐ `make test` byte-exact (no behavior consumes the field yet).
- ☐ `seas_spatial_dyn_driver` builds; `--dry-run` prints the floor (add to the banner).

Dependencies

Depends on: nothing. Required by: Phases 2, 3.

2.5 Phase 2 – apply the floor in the LSW strength (the SAFS blow-up path)

Goal

LSW shear strength uses $\max(\sigma_n, \sigma_{n_floor})$; default-disabled is bit-identical to today; enabled (10 MPa) caps the strength so a tensile σ_n no longer free-slides.

Files to Modify

- `dynamic/tpv205_friction.hpp` — `SolveLSW_TPV205(...)`: add a **trailing defaulted** parameter `real_t sigma_n_floor = 0.0` (after `tau2_corr`). Replace line 157 `const real_t sigma_n_pos = std::max<real_t>(sigma_n_total, 0.0);` with `const real_t sigma_n_pos = std::max<real_t>(sigma_n_total, sigma_n_floor);`. Default `0.0` $\Rightarrow \max(\sigma_n, 0) \Rightarrow$ byte-exact for every caller that omits it.
- `dynamic/fault_face_flux.cpp`:804 and :921 — pass `sigma_n_floor_arg` (see below) as the new trailing arg to `SolveLSW_TPV205`.
- `dynamic/tpv205_substep_iterator.cpp`:110 — pass `flux_.SigmaNStrengthFloorForLSW()` as the new trailing arg (the iterator holds `FaultFaceFlux& flux_`).
- `dynamic/fault_face_flux.hpp` — add accessor `real_t SigmaNStrengthFloorForLSW()` `const { return sigma_n_strength_floor_ >= 0.0 ? sigma_n_strength_floor_ : 0.0; }`

(maps the disabled sentinel $-1 \rightarrow 0.0$, i.e. current LSW $\max(\sigma_n, 0)$).

Interfaces

- `SolveLSW_TPV205(..., real_t &tau1_corr, real_t &tau2_corr, real_t sigma_n_floor = 0.0).`
- `FaultFaceFlux::SigmaNStrengthFloorForLSW() const.`

Edge Cases

- floor disabled (-1) $\rightarrow$ accessor returns $0.0 \rightarrow \max(\sigma_n, 0) \rightarrow$ byte-exact.
- floor = 10 MPa, σ_n tensile (-2.9 GPa) $\rightarrow \sigma_{n_pos} = 10$ MPa, $\tau_{strength} = \mu_{eff} \cdot 10$ MPa (finite) $\rightarrow V_{abs} = (|\tau| - \mu_{eff} \cdot 10 \text{ MPa}) / \eta_s$ is bounded (vs free slip $|\tau| / \eta_s$).
- the `mu_eff >= 0.5 * mu_s_barrier` lock (line 117) and `SEAS_NOOPENING` (line 144) branches are **before** line 157 and are unchanged.

Acceptance Criteria

- ☐ New unit test `test_lsw_strength_floor` on `SolveLSW_TPV205` directly: with $\sigma_n = -2.9e9$, $\mu_{eff} = 0.3$, $\eta_s = 4.6e6$, $|\tau| = 1.7e9$: floor $0.0 \rightarrow \tau_{strength} = 0$, $V_{abs} = |\tau| / \eta_s$ (free slip); floor $10e6 \rightarrow \tau_{strength} = 0.3 \cdot 10e6 = 3e6$, $V_{abs} = (|\tau| - 3e6) / \eta_s$ (bounded, $\approx$ free-slip minus a small finite cap) — assert the floored $V_{abs} <$ free-slip V_{abs} and $\tau_{strength} == \mu_{eff} \cdot \sigma_{n_floor}$.
- ☐ `seas_test_*` for TPV205 / the substep-iterator parity **byte-exact** with floor disabled (the default- 0.0 arg path).
- ☐ make test green.

Dependencies

Depends on: Phase 1. Required by: Phase 4.

2.6 Phase 3 – apply the floor in the RS strength (aging path), gated

Goal

RS shear strength uses $\max(\sigma_n, \sigma_{n_floor})$ when enabled; disabled keeps $\text{abs}(\sigma_n)$ (byte-exact TPV102/104).

Files to Modify

- `dynamic/fault_face_flux.cpp:226` and `:562` — replace `std::abs(s.sigma_n_total) / std::abs(sigma_n_fric)` with a helper:

```
const real_t sn_str = (sigma_n_strength_floor_ >= 0.0)
    ? std::max(s.sigma_n_total, sigma_n_strength_floor_) // (sigma_n_fr
    : std::abs(s.sigma_n_total);
real_t strength = sn_str * f_V;
```

Note the friction *solver* call (`solver.Solve(..., std::abs(s.sigma_n_total), ...)` at `:201/:548`) also takes σ_n — **decide in /code-review** whether to floor that argument too (it sets the Brent bracket / Theta); v1 floors only the explicit strength to minimize the byte-exact surface.

Edge Cases

- disabled (-1) $\rightarrow \text{abs}(\sigma_n)$ (current). enabled $\rightarrow \max(\sigma_n, \text{floor})$.
- TPV102/104 σ_n is compressive ~ 120 MPa $\Rightarrow$ with no floor set, $\text{abs} \equiv \text{current} \Rightarrow \text{byte-exact}$.

Acceptance Criteria

- ☐ seas_test_tpv102_nuc_callback_parity, test_resolve_rate_state_guards, the RS np=2 test: **byte-exact** with floor disabled.
- ☐ Unit: RS strength at $\sigma_n = -1\text{e}9$ with floor $10\text{e}6 \rightarrow \text{strength} = f_v \cdot 10\text{e}6$ (bounded), not $f_v \cdot 1\text{e}9$.

Dependencies

Depends on: Phase 1. Required by: Phase 4.

2.7 Phase 4 – enable on the SAFS configs + blow-up acceptance

Files to Modify

- safs/.../config/spatial_friction_slip_weakening_safs_projected_stress_resolution_Dc2.toml (and Dc8/_Dc10, and the RS configs) — add under [friction]: $\text{sigma_n_strength_floor_pa} = 10.0\text{e}6$ # cohesion-like floor: below 10 MPa compression, $\text{strength} = \mu \cdot 10\text{MPa}$ (constant).

Acceptance Criteria

- ☐ --dry-run banner reports sigma_n strength floor: 10 MPa.
- ☐ **Blow-up arrest (cluster, the real test):** rerun the LSW Dc2 case (same as job 7748818, mixed_flux=adjacent) with the floor at 10 MPa to $\sim t=35$ s; assert V_{max} stays bounded (e.g. $< 1\text{e}3$ m/s) past the original $t=28$ s onset — i.e. the free-slip runaway is broken. Run **independently** of the mixed_flux=none test so we learn whether the floor alone suffices, the flux alone suffices, or both are needed.
- ☐ make test green (the four benchmarks byte-exact; floor unset there).

Dependencies

Depends on: Phases 1–3.

2.8 Testing Strategy

- Phase 1: config-parse unit (present/absent/negative).
- Phase 2: direct SolveLSW_TPV205 unit at a tensile σ_n (free-slip vs floored, numbers above); TPV205 byte-exact parity (floor off).
- Phase 3: RS strength unit + TPV102/104 byte-exact parity (floor off).
- Phase 4: cluster rerun (bounded V_{max}) + full make test.
- Cross-cutting: a grep test that the floor reaches **all four** strength sites (Table above) — a missed site silently leaves a free-slip path.

2.9 Risk Assessment

- **Missed strength site** $\rightarrow$ a residual free-slip path; mitigate with the all-sites grep test and the cluster rerun (if it still blows up, a site was missed).

- **Byte-exactness regression** → the default-disabled sentinel + default-0.0 SolveLSW parameter must reproduce current expressions exactly; the parity tests guard this. The SolveLSW_TPV205 signature change touches the oracle kernel — the trailing default is the safety.
- **Symptom vs root (carry from §6b)**: the floor bounds the friction-mediated feedback but the σ_n oscillation source (central-flux / sliver spatial instability) remains; expect a *bounded but locally spurious* solution at the slivers. Pair with `mixed_flux=None` + mesh repair.
- **Floor masks legitimate opening** — fine for this pure strike-slip SAFS case (no opening expected); scope the option off for normal-fault/bimaterial problems.

2.10 Decisions for /code-review

1. **RESOLVED (2026-05-26, user)**: use the **active $\mu_{\text{eff}}(\delta)$** — $\tau_{\text{strength}} = \mu_{\text{eff}}(\delta) \cdot \max(\sigma_n, \sigma_n_{\text{floor}})$ (C^0 -continuous at the threshold, keeps slip-weakening; “constant in σ_n ” below the floor but still follows $\mu(\delta)$). NOT the literal $\sigma_n_{\text{floor}} \cdot \mu_s$ hard constant (which would jump at the threshold). The plan as written already reflects this — no change needed.
2. **Floor the friction-solver σ_n argument** (:201/:548, :226-RS) too, or only the explicit strength? (Affects Theta/bracket; v1 = strength only.)
3. **Scalar (this plan) vs per-DOF floor** (depth-dependent cohesion à la TPV31)? v1 scalar.
4. **Also cap σ_n itself / the radiated normal traction**, not just the strength? (Bounding V may already bound $[[v_n]] \rightarrow \sigma_n$; confirm empirically before adding.)
5. Config key location: `[friction] sigma_n_strength_floor_pa` (top-level, applies to both laws) vs per-law block. Plan = top-level.